

Predicting airline acceptance of fuel-saving reroutes from flight-plan revisions

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Abstract

Flight-plan revisions provide direct comparisons for learning route preferences: a route previously filed for a flight and its replacement. We use 1.48 million pairs from eighteen months of European traffic to train a ranking model with route indicators, flight context and waypoint identifiers. Shared flight attributes prevent message timing from revealing the label. On a later test quarter, the model identifies the replacement in 68.1 % of cases, compared with 55.0 % for the best single-indicator rule and 57.5 % for the strongest logistic baseline. Accuracy reaches 94.9 % on the fifth of pairs with the highest confidence. Matched examples of retained routes improve predictions on a smaller stay-pair test set. A retrospective policy then selects lower-fuel routes above a calibrated probability threshold. At an operating point targeting 80 % precision, 79.2 % of selections match the subsequent filing and account for 20.1 kilotonnes of planned-fuel differences over the quarter. These results establish useful prediction and confidence-based selection from revision records. The fuel total describes historical choices; acceptance of actual proposals and additional savings require prospective evaluation.

Keywords: air traffic flow management, airline preferences, learning to rank, fuel efficiency

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1. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the airline subsequently re-filed. The new route shed that delay completely, and it cost the airline on each of the three counts an airline pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. The records show a substantial trade-off between attributed delay and other route indicators. They do not reveal the airline’s internal costs or all the circumstances behind its decision.

This paper asks whether recorded revisions can be used to predict route choice. An airline may revise a plan after a delay is attributed, an inbound aircraft arrives late, a forecast changes, or a route becomes unavailable. The archive records the earlier route and its replacement, but does not identify which of these reasons prompted the change. We use each pair as an observation of filing behaviour: given the two routes and their characteristics, can a model identify the replacement without being told their filing order?

That question matters because route indicators do not determine choice on their own. A lower-fuel route may carry more delay or higher en-route charges, and the effect of delay depends on the aircraft’s subsequent flights. Published assessments of European network performance also recognise the role of capacity constraints in route extension (Performance Review Commission, 2026). Predicting a route choice is therefore a different task from finding the route with the lowest value of one indicator. Here the aim is to learn patterns in recorded choices, without reconstructing an airline’s private cost function or attributing inefficiency to an individual operator.

The main methodological contribution is the construction of the training comparison. Both routes in a revision pair were filed for the same flight. This provides a direct observation of a replacement, whereas ranking a filed route above generated alternatives assumes a preference that the record may not establish. Revisions also present two challenges: message timing can reveal which route came later, and the sample excludes flights that kept their routes. We address the first through feature construction and study the second using matched examples of retained routes.

The second contribution is an evaluation of prediction reliability. We test on a later period, compare the model with simple rules and learned baselines,

and examine how accuracy changes when low-confidence predictions are withheld. Ablations and feature attributions answer complementary questions: which inputs improve a retrained model, and which inputs influence the predictions of the fitted model?

The third contribution is a retrospective evaluation of selective fuel-saving proposals. It asks whether a confidence threshold identifies lower-fuel routes that agree with subsequent filings, how much planned fuel those matching choices represent, and how the results vary with the threshold and stay-pair weighting. This use case illustrates the model’s potential value. It does not measure responses to actual proposals or additional savings caused by a service.

This manuscript develops the work described in the companion SESAR Innovation Days 2026 conference paper. The conference version presents the revision-based method and its main results. This journal version adds the detailed baseline comparisons, encoding and ablation analysis, proposal-policy evaluation, and discussion of the conditions under which recorded choices may inform future proposals.

The remainder of this manuscript is organised as follows. Section 2 explains why a revision constitutes a preference observation and why it carries a stronger label than generated candidates; Section 3 positions the work in the literature. Section 4 describes the archive and pair construction, Section 5 the model and its calibration, Section 6 predictive performance, and Section 7 the proposal-channel simulation. Section 8 collects the threats to validity, and Section 9 concludes.

2. Background

This section introduces the filing process, explains what a revision label records, and distinguishes route-choice prediction from its potential uses. It also relates the pairwise model to established choice-modelling methods.

2.1. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised at any time until the flight leaves. Every filing is a message, and so is every later change, and the network keeps them all, so that what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own:

the message that changed the horizontal route of the plan, and the message that preceded it.

The two messages of a pair together describe a decision, because the airline had committed to the earlier route and then filed the later one for the same flight on the same day, which makes the later route the one it chose and the earlier one the route it abandoned. That the pair is assembled from two messages sent at different times is not a detail of bookkeeping: it shapes the encoding, and Section 4.3 returns to it.

The sharpest of those decisions, and the ones in which the terms of the trade are clearest, are made under flow management. When expected traffic exceeds the capacity available in a volume of airspace, a *regulation* limits the rate at which flights may enter it. The imbalance can reflect demand or capacity changes associated with staffing, weather or equipment. Flights planning to cross the volume receive calculated take-off times consistent with that rate. If a calculated take-off time is later than the requested time, the difference is the flight's *attributed ATFM delay*. The aircraft waits on the ground rather than in the air. Not every flight crossing a regulated volume is delayed, and a flight affected by several regulations is attributed to the most penalising one.

An airline facing such a delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel, en-route charges and the departure slot it already holds, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation (i.e., a schedule in which the same aircraft flies several legs in the day, so that delay on one leg propagates to the next) may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. The decision is worth learning rather than deriving, since the trade depends on airline-specific and flight-specific costs that the network does not observe: published European reference values (Cook and Tanner, 2015) give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular airline applies, and what an individual operator would charge itself for forty minutes is not known outside it.

Most revisions in this archive have no recorded regulation: in seven pairs in ten neither route carries a regulation, and Section 4 reports what that majority looks like. A dispatcher re-planning after a forecast update, or after the aircraft's earlier legs have moved, leaves the same kind of trace as one escaping a delay, and the archive records what changed rather than why.

2.2. Potential uses of the predictions

One potential use is to screen fuel-saving route proposals. For a flight with a filed route and a feasible alternative, a preference model could help identify when the lower-fuel option is also likely to be chosen. A confidence threshold would allow the system to withhold uncertain suggestions. Section 7 examines that principle on historical revision pairs, using subsequent filings as the reference.

A second research question concerns voluntary rerouting when demand exceeds the capacity of an airspace volume. Predicted preferences might help compare which candidate reroutes are both acceptable to operators and useful for reducing demand. That question would require evaluating route choice together with network effects; the present study contains no overload-release experiment.

For either use, agreement with a historical filing is evidence about choice prediction, not a measured probability of accepting an unsolicited proposal. Actual responses would provide a more direct evaluation. They would also introduce selection bias: only the proposals sent could receive responses, and confidence filtering would favour proposals predicted to succeed.

2.3. Revealed preference and the construction of pairs

Readers of this journal will recognise the setting as a route-choice problem, a family with a long history in travel-demand modelling (McFadden, 1974; Ben-Akiva and Lerman, 1985; Train, 2009). The observation unit here is the choice between exactly two alternatives that the decision maker itself constructed, which avoids generating alternatives for the training labels, and the estimator is a gradient-boosted ranker rather than a parametric logit.

The pairwise loss has the same logistic form as a binary choice model applied to the difference between two utility scores. Here the score is learned with decision trees rather than a linear combination of inputs. This allows interactions between route characteristics and flight context, but the fitted scores are predictive quantities: they do not identify an airline’s economic utility or willingness to pay.

The connection to revealed preference is the use of observed behaviour rather than stated intentions (Samuelson, 1938). For a revision, the observed behaviour is precise: the airline replaced one filed route with another. Interpreting this as a preference between simultaneously available options is an additional assumption, because conditions may have changed between filings.

Stay pairs, introduced in Section 4.2, require a further assumption because their alternatives come from different flights.

The label construction determines which comparison the model learns. The authors’ earlier route-ranking work aimed to predict plans not yet filed, allowing traffic demand to be estimated before the filing horizon (Dalmau et al., 2024). It ranked each filed plan above generated alternatives, using up to eleven route options from the Network Manager, EUROCONTROL’s network-level flow-management function. On the pre-tactical set, none of those alternatives need match the eventual filing.

With generated alternatives, successful prediction can partly reflect differences between filed and generated routes. The archive alone does not tell us whether the airline examined and rejected those alternatives. This does not make generated routes unsuitable as candidates; it limits what can be inferred when they are labelled as rejected choices.

A revision offers a more direct comparison (Fig. 1). The airline filed both routes, one after the other, for the same flight. The label therefore records a replacement that actually occurred. It does not prove that both routes appeared on a dispatcher’s screen or remained available at the same decision time. No dataset in this study provides both label constructions for the same flights, so their relative predictive performance is not measured here.

The model is trained and evaluated on pairs. In principle, a scoring function could also compare several candidates for one flight. However, the indicators are normalised relative to the alternatives being scored (Section 4.3). Adding a candidate can change those reference values and therefore the scores. Extending the method to larger sets requires evaluating that dependence and checking whether the pairwise calibration remains appropriate; the present results establish neither.

The price to be paid is that revisions are not a random sample of flights, but rather flights that changed, which introduces a bias of its own. Rather than assuming that bias away, Section 6.3 measures it.

3. Related work

Three threads of literature meet in this paper, and the first, rerouting in air traffic flow management, has largely approached the problem from the network side. Rerouting was formulated as a dynamic network flow problem (Bertsimas and Patterson, 2000) and later as a stochastic extension that

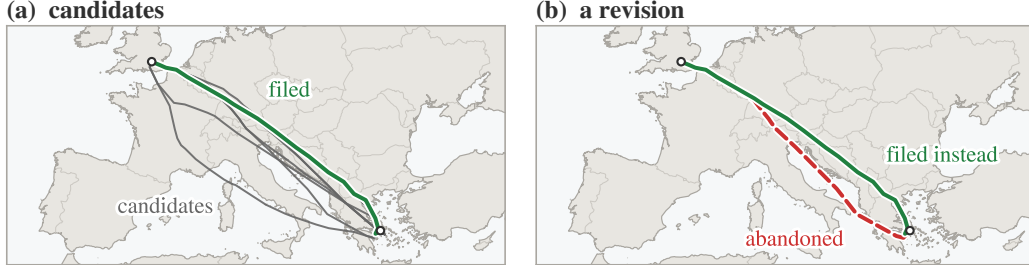


Figure 1: Two ways to build a training pair, on one city pair; each route is labelled by its role where it is drawn. Panel (a): the filed route (solid green) is ranked above candidates from a route generator (grey), which the airline may never have evaluated; those drawn here are other routes observed on the same city pair. Panel (b): a revision supplies two routes that the same airline filed for the same flight hours apart, one of which it abandoned. The right-hand pair directly records which filed route replaced the other.

reroutes flights as capacity evolves (Mukherjee and Hansen, 2009); both optimise a network objective and assume the resulting assignment is executed, so the airline’s willingness to fly it is not modelled.

The value of airline collaboration in flow management has been examined (Castelli et al., 2012), which motivates the channel simulated here but not which proposals would be taken up. Closer still, incorporating airline preferences into collaborative rerouting changes which reroutes are selected (Murça, 2018; Xu et al., 2020), as does the collaborative trajectory options programme, where an airline attaches to each alternative a cost relative to its preferred option (Kulkarni, 2018). In all of these the preference is an input the airline must supply; this paper learns the preference function itself, at network scale, from what airlines did rather than stated, providing a predictive estimate that could complement explicitly supplied preferences.

Other studies estimate preferences from observations rather than asking airlines to supply them. Historical trajectories have been clustered and classified to provide options for demand-capacity optimisation (De Giovanni et al., 2024). The probability of a regulated flight rerouting on its own has also been predicted to help assess the consequences of a regulation (Dalmau, 2022). Those tasks concern behaviour without a proposal. Predicting the response to a particular proposed route is a related but distinct question.

Airlines also express priorities within a regulation through the user-driven prioritisation process (Pilon et al., 2021; Ruiz et al., 2019). This orders delay among an airline’s own flights rather than choosing routes. Explicitly

supplied priorities and predicted route preferences could therefore provide complementary information.

The second thread is economic: earlier than any of the above, route choice in Europe was studied as a trade between fuel and en-route charges (Delgado, 2015), documenting economic influences on choice without using revision pairs, and the discrete-choice tradition of Section 2.3 supplies the theoretical frame for such labels; revisions provide a useful source of paired filing observations, although simultaneous availability of both routes is not established.

The third thread is methodological: learning a ranking function from pairs is standard (Burges et al., 2005; Liu, 2009), and learning preferences from logged behaviour rather than elicited judgements was established in information retrieval, where clicks on search results yield reliable relative, though not absolute, relevance labels (Joachims, 2002). The revision archive is the aviation analogue of a click log, in that nobody was asked anything and each record must be interpreted in the context in which it arose; the same literature supplies the caution this paper inherits, that logged data reflects the behaviour of the system that logged it (Section 8). The direct predecessor of this work (Dalmau et al., 2024) established the framing adopted here, a gradient-boosted ranker scoring routes by their indicators in context, and reported the difficulty motivating the present paper: its training pairs rank the filed plan above candidates the airline may never have examined, the argument and label construction Section 2.3 sets out in full.

4. Data and pair construction

This section describes the archive, pair construction and feature encoding. Section 5 then explains model training and calibration.

4.1. *The archive*

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. Each pair is one training example containing two route rows and a label identifying the replacement. Planned fuel refers throughout to the fuel estimate attached to the recorded plan, not measured fuel burn. Each route is described by twenty-three features, summarised in Table 1, one of which is in-house shorthand

Table 1: What the model sees for each route; some rows summarise several features (twenty-three in all). The first two groups and the last belong to the route and differ within a pair, the indicators entering as differences from the pair minimum; the rotation and context rows belong to the flight and carry the same value on both routes, the one exception being the outbound connection time, which is shortened on the slower route by the difference in flight time and is therefore route-dependent after all.

group	content
indicators	flight time, distance, planned fuel, en-route charges (the fees an airspace user pays for crossing each charging zone)
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume
rotation	inbound and outbound connection times and states
context	airline, aircraft type, city pair, hour, weekday, month, time to departure
route	waypoint identifiers, encoded as unordered tokens

and worth spelling out: the protected volume is the volume of airspace whose capacity the regulation was declared to protect.

“Airline” is used loosely throughout for any aircraft operator, since the archive contains business, cargo and non-scheduled traffic as well as scheduled airlines, and the figures say “operator” where the count is over all of them. All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper. A small number of individual flights are reported as worked examples, and nothing beyond what each example needs is given about them.

What those 1.48 million decisions look like matters before any model touches them, and Table 2 breaks the archive down by the regulation context in which each revision was made.

Seven revisions in ten carry no regulation on either route. Their median planned-fuel difference is one kilogramme, although that median does not establish why individual plans changed. Among revisions that leave a regulation, the replacement uses a median 101 kg more planned fuel and 2.1 minutes more flight time. Only 28.1 % reduce planned fuel.

Table 2: The revision archive by decision context, over all eighteen months. A route counts as regulated when a regulation was attributed to the flight while filed on that route (the delay itself is free to be zero; Table 4 sorts the held-out quarter by the delay instead, giving different counts). Unregulated pairs carry no regulation on either route, pairs leaving a regulation carry one on the abandoned route only, the next two carry one on both, and the last only on the newly filed route; the median fuel difference is between the two routes of a pair, and the final column is the share where the newly filed route burns less.

context	pairs	share [%]	median Δ fuel (new – old) [kg]	new route saves fuel [%]
unregulated	1 047 228	70.6	1	46.2
left regulation	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
became regulated	21 337	1.4	–5	51.8

The smallest group, 1.4 % of revisions, moves onto a route with a regulation. It is also the only group in this table in which more than half of the replacements use less planned fuel. These observations show why the lower-fuel route need not be the route filed next. They do not identify the full costs or reasons behind each change. Two examples appear in Fig. 2.

4.2. *Stay pairs*

Revision pairs describe only flights that changed route. They do not record the alternatives considered by flights that kept their route. A model trained on revisions can therefore learn a pattern that is useful within that sample but favours changes too readily elsewhere. Section 6.3 measures this effect on a smaller, separately constructed set of *stay pairs*.

Each stay pair joins two flights on the same day, with the same origin, destination and aircraft type. They flew different routes and neither revised its plan. One flight was regulated and the other was not. We label the regulated flight’s retained route as preferred and use the other flight’s route as its comparison alternative. Both route rows are given the retained-route flight’s context before scoring. The model therefore compares routes for that flight, not the two flights’ scores.

The retained route is observed; rejection of the comparison alternative is inferred. The other flight demonstrates that its route was used that day, but not that the regulated flight considered it or could have used it at its own departure time. The label is consequently weaker than a revision label.

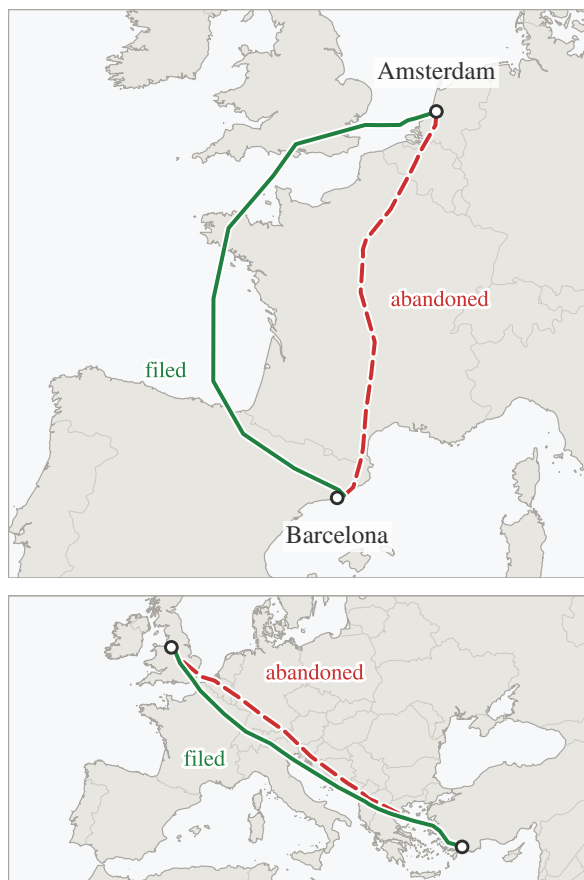


Figure 2: Two revisions from the archive, in the visual language used throughout this paper: the abandoned route is dashed red and its replacement solid green, each labelled on the track itself. Top: an escape, in which the airline paid fuel and time to leave a regulated volume; the volume itself is not drawn, since naming the airspace would identify it, and the two airports are named because this city pair is the running example of Section 1. Bottom: a fuel saver, in which the replacement required less planned fuel; its endpoints are left unnamed, that running example being the only city pair this paper identifies. The first kind accounts for a quarter of the regulated archive; the second is what a channel of this kind exists to surface earlier.

In particular, it cannot distinguish preference for the retained route from differences in availability or unrecorded constraints.

There are 6672 stay pairs, including 1093 in the test months, compared with 1.48 million revision pairs. Stay pairs are split by time and added to the

corresponding revision subsets. During training, an increased group weight gives each stay pair a larger contribution to the ranking loss.

The flights that stayed faced a median peak attributed delay of 31 minutes. In 47.5% of stay pairs, the retained route uses more planned fuel than the comparison route; the median difference is 3 kg in favour of the retained route. These figures describe retained routes despite delay and similar planned fuel. They do not, by themselves, demonstrate a preference for stability or rejection of an available alternative.

One property of this construction has to be stated alongside it, because it bounds what the stay pairs can ever be used to measure. The regulated flight is always the one whose route was kept, so regulation status identifies the labelled answer. Concretely, in all 1 093 stay pairs of the test months the kept route carries a positive attributed delay and at least one regulation, and the alternative carries neither, with no exceptions. A rule as crude as “prefer the route of the flight that was regulated” is therefore right on every stay pair without examining a route at all. Whether the model takes that shortcut is a question the archive can answer, because it also contains the mirror population in which the same rule is always wrong; Section 6.3 puts the two side by side.

4.3. Encoding and leakage control

The replacement route comes from the later message. If message timing were allowed to differ between the two rows, it could reveal the label without providing any information about route choice. This is called *label leakage*: the input contains a clue to the answer that would not be available when predicting a new decision.

One held-out pair illustrates the problem. The original plan was filed at 00:30 and its replacement at 10:56 for a flight due off the blocks around 12:30. Time to departure would be 720 minutes for the earlier message and 94 minutes for the later one. A rule choosing the smaller value would identify the replacement without comparing the routes. Across the test quarter, the median interval between filings is 33 minutes, a quarter of pairs are more than 90 minutes apart, and the replacement is the later message in all but a thousandth of pairs. The same shortcut would persist in a chronological test set.

We therefore give both rows the same flight-level context, taken from the later message; time to departure takes the pair minimum. For stay pairs, the context of the flight that retained its route is copied to both rows.

Route-specific quantities can still differ. One deliberate exception within the rotation features is outbound connection time: it is shortened for the slower route by the difference in flight time. Aligning context removes the direct timing shortcut. It does not reconstruct both routes’ indicators at a common historical decision instant; that distinction matters for prospective use.

The four route indicators are then expressed relative to the smaller value in their pair. For indicator k and route $r \in \{A, B\}$,

$$\tilde{x}_{rk} = x_{rk} - \min(x_{Ak}, x_{Bk}). \quad (1)$$

For example, planned-fuel values of 2 000 and 2 300 kg become 0 and 300 kg. The operation is applied separately to fuel, flight time, distance and charges; the minimum for each indicator need not belong to the same route. Units remain unchanged. A zero fuel input means no extra fuel relative to the comparison route, not zero fuel for the flight. The encoding emphasises the extra requirement of one route relative to the other, while shared context still tells the model which flight is being considered.

This normalisation is a representation choice, not another leakage correction. Absolute fuel use is valid information, but the comparison encoding performs substantially better in the ablation reported in Section 6.3. It does not impose a common economic value on fuel differences.

4.4. *Splits*

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720), the held-out quarter cited throughout; the three account for all 1.48 million revision pairs of the archive. These are pair counts: a flight may contribute several revisions. Stay pairs are assigned to the corresponding periods and evaluated separately. Every threshold, calibrator and stopping decision is taken on the tuning month alone, flights appearing in the tuning or evaluation months are removed from the fitting set, and eleven automated checks verify split disjointness, feature construction and the pair invariants.

5. Model, training and calibration

This section separates three operations: learning route scores, turning a score difference into a probability, and choosing a threshold for selective use.

The model follows the predecessor system (Dalmau et al., 2024), with the encoding and stay-pair construction described above.

The model assigns each route a score $s(x)$ from its encoded features x . The route with the higher score is the predicted choice. Scores have no physical units and are not probabilities. Training uses pairs with one route labelled preferred. If their scores are s_+ and s_- , the pairwise logistic loss is

$$\ell = \log(1 + \exp[-(s_+ - s_-)]) . \quad (2)$$

The loss is large when the labelled preferred route scores lower. It decreases smoothly as that route’s lead grows, encouraging both the correct ordering and a larger score gap (Burges et al., 2005). A pair’s group weight multiplies its contribution to training. For revisions, s_+ is the replacement’s score; for stay pairs, it is the retained route’s score. The plus sign identifies the training label, so s_+ can be lower than s_- when the prediction is wrong.

The scoring function uses gradient-boosted decision trees in CatBoost (Prokhorenkova et al., 2017), with depth 8, an automatically selected learning rate and early stopping on the tuning month. Early stopping uses the tuning criterion to select how many trees to retain, rather than continuing training indefinitely. Each successive tree helps improve the current scores. The trees can represent interactions: for example, the effect of a route’s extra flight time can depend on the time available before the aircraft’s next flight. Stay-pair weights of 10, 30 and 100 are examined; the adopted value is $w = 10$ (Section 6.3).

The route’s waypoint identifiers are encoded as a bag of tokens. Each identifier is treated as a word, and the model receives the collection of words used by the route. Order and coordinates are discarded. This representation lets two routes with similar aggregate indicators differ according to the published points they use. It can learn associations with those parts of the network, but it does not explicitly reconstruct the route’s geometry or airspace crossings.

After training, the absolute score gap $g = |s_A - s_B|$ measures confidence in the ordering. Calibration fits a mapping $q(g)$ from that gap to the probability that the higher-scoring route matches the label. A value of 0.8 means that about eight in ten comparable predictions are expected to be correct. It is a statement about a group of predictions, not a promise about one flight. To fit the mapping, we score the tuning pairs and compare each ordering with its known label. Their score gaps and correctness indicators provide the

calibration data. The mapping is then fixed for the test quarter (Zadrozny and Elkan, 2002); fitting it does not change the underlying route scores.

For the worked example, scores of -2.82 and $+0.90$ give a gap of about 3.7, which maps to 0.99. In that test-set gap band, the model is correct in 97.4% of cases. A gap of about half a point maps to approximately 0.72, compared with 68.5% observed accuracy in that band. These comparisons show what calibration means and how it can overstate confidence on later data.

We compare the raw sigmoid of the gap, Platt scaling (a fitted sigmoid) and isotonic regression (a stepwise fit constrained not to decrease with the gap). Isotonic regression has the lowest tuning-month log loss and is selected before testing. Log loss rewards accurate probabilities and penalises confident mistakes. On the held-out quarter, the raw sigmoid has an expected calibration error of 0.9 percentage points, compared with 1.8 for the selected isotonic mapping. This error averages the discrepancy between predicted probabilities and observed frequencies across fifteen bands; it does not bound every band’s discrepancy. Across the coarser reported gap bands, the largest discrepancy is 4.4 points. Calibration methods can behave differently across periods and evaluation criteria (Guo et al., 2017); the test result is reported without changing the selection made on the tuning month.

The proposal policy uses the probability for the lower-fuel route. If that route has the higher score, its probability is $q(g)$; otherwise it is $1 - q(g)$. A threshold on this quantity determines which pairs are selected. Threshold and precision are distinct: a threshold of 0.600 selects candidate probabilities strictly above that value, whereas precision is the observed fraction correct among all retained predictions. Section 7 selects thresholds for target precision on the tuning month and measures the resulting precision on the test quarter.

6. Predictive performance

We first compare overall accuracy with simpler methods and examine how it changes when the model withholds uncertain predictions. We then report results by decision context, assess the effect of model inputs and stay-pair training, and inspect errors and feature contributions.

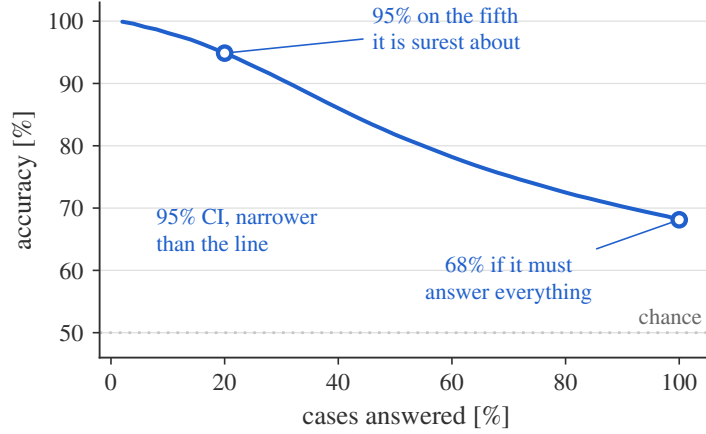


Figure 3: Accuracy against coverage on the held-out quarter. The horizontal axis is the share of cases the model is required to answer, lowest confidence dropped first; the vertical axis is the share of answered cases it gets right. Moving left, the model answers fewer cases and is right more often: 95 % of the fifth it is surest about, against 68 % when it must answer everything. A “case” here is one pair of routes, and answering it means naming one of the two. The dotted line is chance; the 95 % confidence interval is drawn but is narrower than the line itself at every point. It is this curve, rather than any single figure on it, that characterises the system.

6.1. Accuracy, coverage and baselines

On the held-out quarter the model identifies the airline’s chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, or 67.7–68.6 under city-pair clustering), against the 50 % of random choice. The tuning-month accuracy is 67.5 %, close to the test result. The chronological test provides evidence of performance on later data; this comparison alone does not rule out overfitting.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 3 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Here, coverage means the fraction of route pairs on which a prediction is retained. It counts pairs, not distinct flights or proposals. On the fifth with the largest gaps, accuracy reaches 94.9 %. This selection ignores which route uses less fuel. A fuel-saving policy has an additional condition, so its own coverage and precision must be evaluated separately (Section 7).

The curve is one reference; the other is what simpler methods achieve on the same held-out quarter. Table 3 sets out the whole ladder, from a coin

Table 3: Baselines on the held-out quarter, 281 720 pairs; not every single-feature rule that was tested is shown, only those that also anchor later discussion. A single-feature rule decides only where both routes carry a value for its indicator and the two values differ, and is scored as a coin toss on the rest; the “decides” column gives that share. A route under no regulation carries no delay figure rather than a zero, which is why the delay rule decides on fewer pairs than the share on which the revision changed the delay in Table 4. The logistic models are fitted on the same 1 134 000 training pairs as the model of this paper, with the same within-pair encoding, and their regularisation strength is selected on the tuning month.

baseline	inputs	decides [%]	accuracy [%]
coin toss	0	–	50.0
prefer less attributed delay	1	14.6	53.2
logistic, cost indicators	4	–	54.9
logistic, and delay and count	6	–	56.4
logistic, all numeric	14	–	56.4
logistic, and categorical context	22	–	57.5
this paper’s model	23	–	68.1

toss to the model of this paper, so that a reader can see where the accuracy is bought and judge whether the complexity earns its place.

The strongest tested single-indicator rule has 55.0 % accuracy. The model improves on it by 13.1 percentage points. This comparison concerns prediction within the revision archive, not an assessment of the efficiency of airline decisions.

The delay rule illustrates why the scoring convention matters. It is correct on 71.6 % of the 14.6 % of pairs where both delay values are available and unequal. Scoring the remaining pairs as coin tosses gives $0.146 \times 71.6 + 0.854 \times 50 = 53.2$ %. An unregulated route has no delay value in this rule’s input, rather than an explicit zero. The rule therefore excludes some comparisons where one route is regulated and the other is not. Its result should be read under that convention, rather than as the strongest possible delay-only predictor.

Together, these baselines show that the tested single-indicator rules capture only part of the recorded choices. The learned baselines ask whether combining the inputs improves prediction.

A logistic model with the four normalised route indicators reaches 54.9 % accuracy. Adding attributed delay and the regulation count raises this to 56.4 %; eight further numeric features leave it unchanged. Adding categorical context raises accuracy to 57.5 %. This last baseline uses all the inputs except waypoint text.

The gradient-boosted model reaches 68.1 %, a gain of 10.6 percentage points over that logistic baseline. The comparison combines a change in model form with the addition of waypoint text. The separate ablation in Section 6.3 helps distinguish these contributions; the baseline comparison alone cannot assign the whole gain to either one.

The remaining errors also depend on how distinct the routes are. Accuracy is 61.9 % when the routes share more than nine tenths of their waypoints, compared with 74.9 % when they share less than a quarter. Similar routes may provide weaker differences in the recorded inputs. Other influences, such as crew constraints or company planning rules, are not fully represented in those inputs. These are plausible reasons for residual uncertainty, but the archive does not establish an upper bound on achievable accuracy.

6.2. Accuracy by stratum

The 68.1 % headline is an average over decisions of very different kinds, and the first question a reader should ask is what kinds of decision the archive actually contains. Table 4 answers that and the performance question together, on one population: it sorts the held-out quarter by what the revision did to the flight’s attributed ATFM delay, and reports how often the model called each sort correctly.

Seven revisions in ten have no attributed delay on either route. Their median planned-fuel difference is zero, and the model predicts the replacement in 64.0 % of cases. The archive does not establish that these changes are minor: other differences or constraints may matter even when the two recorded fuel values are similar.

Accuracy is higher when a revision removes or reduces delay. Revisions removing delay account for 12.7 % of pairs and are predicted correctly in 88.2 % of cases. Their replacement routes use a median 103 kg more planned fuel; only 26.8 % use less. Revisions reducing delay have a median fuel increase of 56 kg and accuracy of 81.5 %. The remaining contexts, in which delay is unchanged, increased or newly attributed, account for 8.8 % of the quarter and have accuracy near 61 %.

Table 4: What each revision did to the flight’s attributed ATFM delay, on the held-out quarter, and how well the model reads it. Fuel and time differences are the new route minus the one it replaced, so a positive number is a cost the airline accepted; intervals are Wilson 95 %. This table is keyed on the delay itself, whereas Table 2 is keyed on whether a regulation was attributed to a route, and a regulation can be attributed while leaving the delay at zero, which is why 71.0 % of this quarter’s pairs touch no delay against only 63.3 % carrying no regulation (stated with the regulation-exposure strata, below).

what the revision did	pairs	share [%]	median Δ fuel [kg]	new saves fuel [%]	accuracy [%]
no delay either route	199 932	71.0	0	47.7 64.0	[63.8, 64.2]
removed the delay	35 890	12.7	103	26.8 88.2	[87.8, 88.5]
reduced the delay	21 202	7.5	56	31.8 81.5	[81.0, 82.1]
left the delay alone	10 234	3.6	1	45.5 60.4	[59.5, 61.4]
increased the delay	9 189	3.3	−2	52.2 61.0	[60.0, 62.0]
took delay on	5 273	1.9	−10	53.2 60.9	[59.5, 62.2]

These differences show where the recorded choices are easiest to predict. They also explain why overall accuracy is insufficient for a fuel-saving proposal policy: that policy selects lower-fuel routes, which are not necessarily in the most accurately predicted contexts. Section 7 therefore evaluates the selected proposals directly.

One consequence of the archive’s regulation strata is worth stating before the headline is read anywhere else. The held-out quarter is not a random slice of the archive: 63.3 % of it carries no regulation on either route against 70.6 % of the eighteen months, and that unregulated stratum, where accuracy is only 64.0 % (against 88.0 % when only the old route is regulated, 70.6 % when both are, and 58.6 % when only the new one is), has lower accuracy than the two largest regulated strata. Reweighting the four regulation strata to the archive’s own mix gives 67.1 % against the 68.1 % headline, showing that the aggregate result changes by about one percentage point with this change in population weights.

6.3. Ablation and change bias

Table 5 reports screening ablations, each configuration removed in turn and the model retrained, and one of the five dominates the rest.

Within-pair normalisation has the largest effect at the screening setting: removing it reduces accuracy by 10.0 percentage points. This supports presenting the indicators as comparisons between the available options.

Table 5: Screening ablations, each variant retrained from scratch at a deliberately reduced setting (depth 6, 800 boosting rounds; 60.6 % on revision pairs). Values are the change in revision accuracy against that 60.6 % baseline, not the 68.1 % headline model; positive means the variant scores *higher* at that reduced setting. The table reports the size and direction of each change; the full-scale waypoint ablation is reported separately in the text.

variant	Δ accuracy [pp]
without within-pair KPI normalisation	−10.0
without stay-pair correction	+3.7
with monotone constraint	−0.2
without waypoint text	+0.03
including vertical-only revisions	−2.0

The waypoint result changes with model capacity. Removing waypoint text changes screening accuracy by only +0.03 points. At full scale, retaining it improves accuracy by 1.4 points: 68.2 % versus 66.8 %. The 68.2 % result is a separate run of the adopted configuration; its tested seeds span 0.17 percentage points. Thus the small screening model understates the benefit of the text feature. The experiment does not isolate why, but learning from many rare waypoint identifiers may require more capacity.

We also explored a name-independent route representation. A neural encoder received segment descriptors based on bearing changes relative to the great circle and segment lengths relative to total route length. These descriptors remove dependence on absolute scale, rotation and position. Trained with the pairwise objective, the encoder performed worse alone than the token representation and added only a fraction of a point alongside it. A linear classifier on its embeddings separated preferred from abandoned routes only slightly better than chance. The representation may discard useful network context, but these tests do not isolate representation, architecture and optimisation effects. They predate a repair that recovered about a quarter of valid archive pairs, so only their qualitative ordering is reported here.

The stay-pair evaluation asks a different question: does including retained routes improve predictions on that constructed population? With revision-only training, stay-pair accuracy is 34.8 %, so the model favours the comparison alternative in 65.2 % of cases. Adding stay pairs at weight 10 raises accuracy to 90.0 %, while revision accuracy changes from 68.9 % to 68.1 %. The 0.8-point revision difference is measured at full scale; the 3.7-point difference in Table 5 belongs to the reduced screening model.

Regulation status alone cannot explain the two results. In every stay pair, the labelled preferred route is regulated. In the 30 953 test revisions that leave a regulation, the preferred route is unregulated. Always preferring the regulated route would succeed on all stay pairs and fail on all these revisions; reversing the rule would reverse the results. The model reaches 90.0 % and 88.0 %, respectively. It therefore uses information beyond that single rule. This comparison does not rule out other cues associated with how stay pairs were built.

The stay test contains 1 093 pairs, with a 95 % interval of [88.1, 91.7] around its 90.0 % accuracy. Its alternatives belong to different flights and their availability to the retained-route flight is not established. The result is strong evidence of improvement on this test construction, while broader performance on non-revising flights remains to be evaluated.

The weight sweep compares $w \in \{10, 30, 100\}$. Raising the weight from 10 to 100 improves stay-pair accuracy by another 7.3 percentage points and reduces revision accuracy by 1.5 points. The adopted $w = 10$ retains most of the stay-pair improvement while better preserving revision accuracy.

6.4. Error analysis

Fortunately, errors concentrate among low-confidence predictions. Only 5.6 % of all errors occur in the quarter of pairs with the largest score gaps; that share would be 25 % if errors were distributed evenly across confidence levels. Also, 27.7 % of errors are near-ties, defined as gaps smaller than one fourth of the archive’s median gap. These observations explain why withholding low-confidence predictions improves accuracy.

High-confidence errors still occur. In one held-out case, all cost indicators and attributed delay favoured the replacement, but the model ranked the abandoned route higher. This motivates examining route-text effects and interactions. The direction of the inputs alone does not establish which feature caused that particular mistake in a nonlinear model; a local attribution or controlled perturbation would be needed.

Neither these errors nor stay-pair errors are observed rejections of proposals. The 10.0 % stay-pair error rate is an unthresholded ranking error on constructed pairs. It cannot be read as the rate of confident suggestions that an airline would reject. The retrospective proposal policy has its own precision measure, reported in Section 7.

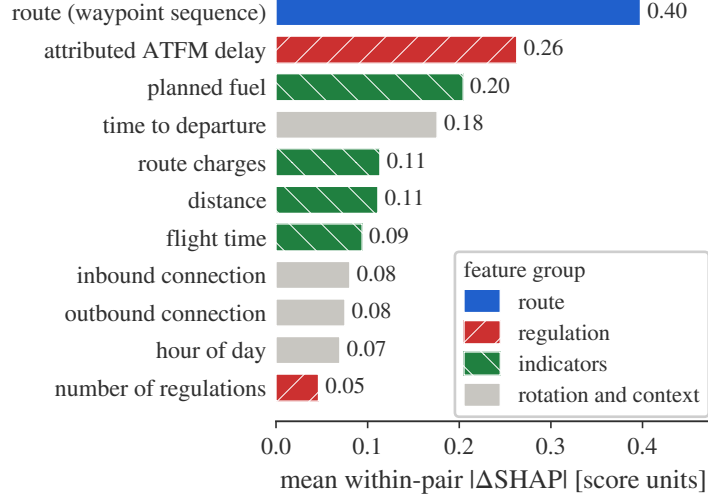


Figure 4: What moves a decision: the mean absolute *difference* between a feature’s Shapley value on the two routes of a pair, over the held-out quarter, in units of the ranking score, with each bar’s value printed beside it. The eleven features that move the score most are shown, of the twenty-three the model sees; fill colour and hatching group them, with the rotation and context groups of Table 1 drawn together. The waypoint sequence is the largest single term, ahead of every individual cost indicator, though the cost indicators taken together sum to more. A large attribution is not the same as a large ablation cost: removing the waypoint sequence and retraining costs 1.4 points of accuracy. The two measure different things, as the text below explains.

6.5. Feature attribution

A ranking model that cannot be interrogated is of little use to an operational user who is asked to trust it. Shapley attributions were therefore computed over the held-out quarter: each feature is credited with the share of the score that is attributable to it, averaged over every order in which the features could have been supplied to the model (Lundberg and Lee, 2017). Because the model decides on a comparison rather than on one route, what is reported here is the *difference* between a feature’s attribution on the two routes of a pair, averaged in magnitude over the quarter (Fig. 4). For example, attributions of +0.3 and +0.1 on the two routes contribute 0.2 to their score difference. These values illustrate the calculation; they are not an additional empirical result. The figure reports the average absolute size of such differences. That is a ranking not of what correlates with acceptance, but of what moves this particular model’s decisions.

Waypoint text has the largest individual mean absolute attribution difference: 0.40 score units, compared with 0.26 for attributed delay and 0.20 for planned fuel. The four cost indicators together sum to 0.52. These values describe how the fitted model distributes its score between inputs; they are not causal effects or economic weights placed on them by airlines.

Time to departure has an attribution of 0.18 even though it is identical for both routes. This is possible because the trees use interactions. For example, a fuel or flight-time difference may affect the score differently close to departure. A shared context feature cannot separate the routes on its own, but it can change the effect of features that do differ. Outbound connection time is different: it varies with route flight time (Table 1).

Attribution direction adds information beyond magnitude. Lower attributed delay contributes strongly towards the chosen route’s score, reaching 2.3 points where that route has one hour and three quarters less delay. Planned-fuel contributions are smaller and spread across more pairs; their tendency to favour higher-fuel routes reflects the fitted relationships in a revision-selected dataset. It should not be interpreted as a general airline preference for burning more fuel.

Removing an individual waypoint identifier and rescoreing changes the score by up to 2.5 points for the most influential identifier. This is a sensitivity test on the encoded route, not a feasible rerouting experiment: deleting a token does not generate a valid alternative. Individual identifiers are withheld in accordance with the study’s neutrality requirements.

Finally, attribution and ablation answer different questions. Attribution describes the fitted model; ablation retrains a model without an input. Although waypoint text has the largest individual attribution, removing it and retraining costs 1.4 accuracy points. Correlated inputs can partly replace its predictive information after retraining. The two results therefore support a useful contribution from route identifiers without implying that they dominate prediction.

7. Proposal channel

This section uses the ranking model to select lower-fuel routes from historical pairs. It evaluates agreement with the recorded replacement and sums planned-fuel differences where that agreement occurs. The purpose is to study confidence-based selection, rather than to estimate the effect of introducing a proposal service.

7.1. Policy and operating points

The policy has three steps. First, keep pairs whose two routes have different planned fuel. Secondly, take the lower-fuel route as the candidate and compute its calibrated probability: $q(g)$ if it has the higher score, and $1 - q(g)$ otherwise. Thirdly, select the candidate only when that probability exceeds τ . Thresholds are chosen on the tuning month for target precision and applied unchanged to the test quarter.

A selection is *confirmed* when the lower-fuel candidate is the route the airline subsequently filed. Precision is the fraction of selected candidates confirmed in this way. Proposal coverage is the fraction of eligible pairs selected. Unlike the accuracy–coverage curve in Fig. 3, this policy always considers the lower-fuel route, which need not be the model’s preferred route. The two selection rules therefore answer different questions. For example, a large score gap favouring higher fuel counts as a confident prediction in Fig. 3 but produces no fuel-saving proposal. A lower-fuel candidate with a more moderate score gap can still pass a fixed probability threshold. Thus the 94.9 % accuracy on the most confident fifth and the 79.2 % proposal precision are not evaluations of the same subset.

Of the 281 720 test pairs, 275 907 are eligible (97.9 %); only equal-fuel pairs are excluded. In 55.1 % of eligible pairs, the lower-fuel route is the route being abandoned. Always selecting the lower-fuel route would therefore match 44.9 % of subsequent filings. The model’s role is to identify the subset where lower fuel and the predicted route choice agree.

Filing order is withheld from the policy. It scores the two historical options without using which is currently filed, so it may select either the replacement or the abandoned route. This differs from a service that only proposes a new alternative to the current plan. It also uses a replacement known from the historical record, rather than discovering that route prospectively. These choices define the evaluation as selection within observed revision pairs. For example, if the lower-fuel route is the later filing and passes the threshold, the selection is confirmed. If the lower-fuel route is the earlier filing and passes, the selection is incorrect, because the airline replaced it. Neither result requires a real proposal to have been sent. The same rule is applied to every eligible pair.

At all reported thresholds ($\tau \geq 0.5$), the selected lower-fuel route must also be the model’s higher-scoring route: the lower-scoring route has probability no greater than one half. Increasing the target precision from 60 % to 90 % increases test precision from 65.8 % to 88.2 %. The share of eligible pairs

selected falls to less than a third of its earlier value, while confirmed planned fuel decreases by 43 %, from 24.3 to 13.8 kt. The selected higher-confidence subset therefore retains a substantial share of the planned-fuel differences.

At the central operating point, $\tau = 0.600$ was selected to target 80 % precision. It selects 56 162 pairs over the test quarter, about one fifth of eligible pairs. Of these selections, 79.2 % match the subsequent filing. For those matches only, we sum the higher planned-fuel value minus the lower one. The total is 20.1 kt, or 63.5 kt of CO₂ using 3.16 kg of CO₂ per kilogramme of fuel. The mean difference is 452 kg per confirmed selection and the median is 130 kg. The distribution is skewed: the largest tenth accounts for 64 % of the total.

These totals concern changes airlines made without receiving proposals. They quantify the planned-fuel differences associated with choices the policy correctly identifies, not additional fuel saved by the policy. They also concern route pairs, not necessarily unique flights; several revisions may belong to the same flight.

Two illustrative annualisations give 70 to 80 kt of planned fuel, or 222 to 254 kt of CO₂. The lower figure applies the per-eligible-pair rate to an annual eligible volume estimated from the eighteen-month archive. The upper figure multiplies the quarter by four. At the 60 % target the corresponding planned-fuel range is 85 to 97 kt. These ranges express two scaling assumptions, not confidence intervals or forecasts of additional savings.

7.2. Concentration of the value, and its sensitivity to the bias correction

The confirmed planned-fuel differences are unevenly distributed. Among the 882 operators associated with at least one selection, 6 account for half of the total and 28 for four fifths. By city pair, 153 of 10 939 pairs account for half. Each test month contributes between 5.9 and 7.5 kt. These are descriptive properties of the sample and selected historical revisions. They do not rank operator efficiency, identify avoidable fuel burn, or prescribe which operators should receive a service.

We also repeat the policy evaluation without stay-pair training. At $\tau = 0.500$, confirmed planned fuel is 24.38 kt without the correction and 24.33 kt with it; precision is 64.4 % and 65.8 %, respectively. At $\tau = 0.600$, the totals are 20.101 and 20.097 kt, with precision of 81.5 % and 79.2 %. Thus the planned-fuel totals change by less than fifty tonnes at either threshold, although precision changes by 1.4 to 2.3 percentage points. The total planned-fuel difference is stable to this training choice within the revision sample.

This does not establish its effect on the broader population of flights that never revise.

7.3. A worked example

Figure 5 follows one retrospective selection. For an unregulated narrow-body flight, the earlier route has planned fuel of 10 485 kg and the replacement 8 469 kg, a difference of 2 016 kg. Flight time is almost unchanged, while route charges increase by 227 EUR. The model scores the routes -2.82 and $+0.90$, respectively. Their gap of about 3.7 maps to probability 0.99, above the fixed threshold $\tau = 0.600$, so the lower-fuel route is selected. It matches the recorded replacement and its planned-fuel difference contributes to the confirmed total.

This large difference is illustrative rather than typical: the median confirmed selection represents 130 kg. The archive does not contain the vertical profiles needed to explain the large fuel difference at similar flight time. The example demonstrates the calculation from scores to a selected route; it does not establish that a proposal would have been accepted or that the full planned difference would have appeared in actual fuel burn.

8. Threats to validity

The results establish predictive value on recorded route pairs. The following questions determine how far that evidence can be extended.

Recorded choice and preference. A revision records the output of a filing process, which combines operator objectives, planning software, available information and constraints. If the earlier route becomes unavailable, replacement is necessary even if it would otherwise be preferred. The archive does not identify that share of revisions. Thus the model predicts recorded choices; it does not separately recover economic utility in the sense of Samuelson (1938). Studies with availability information could separate voluntary substitutions from required ones.

Flights that retain their routes. Stay-pair training raises accuracy from 34.8 % to 90.0 % on 1 093 held-out constructed pairs. Together with 88.0 % accuracy on revisions leaving a regulation, this rules out a fixed rule based only on regulation status. It does not eliminate all construction artefacts. The alternative belongs to another flight, and its availability and consideration are unknown. A stronger dataset would record alternatives available to the



	<i>abandoned</i>	<i>filed</i>
planned fuel [kg]	10 485	8 469
flight time [min]	259	258
route charges [EUR]	2 269	2 496
model score	-2.82	+0.90

$$\text{gap} = 3.7 \longrightarrow P = 0.99 > \tau = 0.600 \longrightarrow \text{proposed}$$

Figure 5: The worked example from the text, on an anonymised city pair. The route initially on file (dashed red) and the fuel-saving alternative (solid green) are shown. The alternative saves 2 016 kg of planned fuel at 227 EUR more in route charges and the same flight time. The calibrated probability exceeds the threshold; the airline’s own later filing chose the proposed route.

same flight and include retained routes across different regulation contexts. This would also help distinguish retention by preference from retention due to other constraints or lack of reconsideration.

Candidate availability and timing. The evaluation obtains the alternative from the subsequent filing. A prospective system would have to obtain it from another source and check its feasibility at the decision time. Historical routes and generated routes are both possible sources, but neither guarantees current availability or the same distribution of inputs seen during training. Harmonising flight context removes a direct timing clue; it does not reconstruct what all indicators would have been when both options were simultaneously assessed. In particular, the alternative’s attributed delay is observed here after it was filed. Before filing, that delay would need to be estimated against the current regulation state. Errors in that estimate could affect the predicted preference.

Route representation and transfer. The waypoint encoding retains identifiers but omits their order and coordinates. The 1.4-point full-scale ablation

measures their contribution in this dataset, not how much would transfer to another region or a changed route network. The initial geometric-encoder experiments did not improve the adopted representation. Further experiments could separate the value of geometry, connectivity and regional context, with evaluation across time and networks. Larger candidate sets also need separate validation because they change the reference values used in normalisation.

Planned fuel and additional savings. The 20.1 kt total sums planned-fuel differences on correctly selected historical revisions. Actual fuel burn can differ because of winds, cruise levels, tactical routing and other operating conditions. The total is therefore neither a measured saving nor a proven upper bound on one. Matching plans to flown trajectories would test how closely the planned differences correspond to actual fuel differences. Separately, those revisions occurred without the policy. Estimating additional benefit requires a prospective comparison that can establish what would have happened without a proposal.

Changes in prices and behaviour. Fuel enters the model in kilogrammes and route charges in euros. Their relative economic importance may change with fuel prices and other conditions. Price-aware features are a testable extension; converting units alone would not guarantee stable preferences. Evaluation over longer periods would also help distinguish changes in input distributions from changes in filing behaviour.

Observing proposal responses. A prospective study could record feasible candidates, the information available when they are scored, and operator responses to proposals. It should distinguish predictive accuracy, proposal acceptance and incremental benefit as separate outcomes. Confidence-based selection also determines which responses are observed, so evaluating the resulting feedback requires accounting for that selection. These questions follow directly from the promising retrospective results and define the evidence needed for a different decision setting.

9. Conclusions

Flight-plan revisions provide useful comparisons for learning airline route-choice behaviour. Their central advantage is the observational link between the two alternatives: both were filed for the same flight. After removing

direct filing-time clues, the model identifies the replacement in 68.1 % of held-out revisions. This exceeds the best tested single-indicator rule at 55.0 % and the strongest logistic baseline at 57.5 %. Accuracy rises to 94.9 % on the fifth of pairs with the highest confidence, showing that the model can also identify predictions that are more reliable.

The experiments clarify which design choices support that result. Within-pair indicator normalisation has the largest effect in the screening ablation. Adding stay pairs raises accuracy on the constructed retained-route test from 34.8 % to 90.0 %, with a 0.8-point reduction on revisions. Waypoint text provides a further 1.4 points at full scale. Its high feature attribution and smaller ablation cost describe complementary aspects of a model whose inputs share predictive information.

The retrospective proposal policy demonstrates selective use of those predictions. At a threshold chosen to target 80 % precision, 79.2 % of selected lower-fuel routes match the subsequent filing, compared with 44.9 % if the lower-fuel option is always selected. The matching choices account for 20.1 kt of planned-fuel differences over the quarter. That total is stable to the stay-pair training choice at the tested thresholds. It describes historical choices successfully identified by the model, rather than savings caused by new proposals.

The next scientific steps are to strengthen the observations for retained routes, test candidates available at a common decision time, and examine representations that generalise across route networks and changing operating conditions. Matching planned and flown trajectories would clarify the fuel interpretation. A prospective study of operator responses would then assess whether the predictive information found in revision records transfers to actual proposal acceptance.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

CRedit authorship contribution statement

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David Perez: Conceptualization, Resources, Validation, Writing – review & editing. **Franck Ballerini:** Validation, Writing – review & editing. **Seddik Belkoura:** Methodology, Validation, Writing – review & editing. **Vincent Taverniers:** Resources, Data curation, Writing – review & editing. **Gratiel Marius Marin:** Resources, Validation, Writing – review & editing. **Robin Deransy:** Conceptualization, Project administration, Writing – review & editing. **Benjamin Cramet:** Supervision, Writing – review & editing. **Grigori Gustin:** Supervision, Project administration, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) and ChatGPT (OpenAI) to assist with drafting and editing the manuscript text and with producing the analysis figures from the study’s result tables. Every analysis figure is drawn by the authors’ own code from the study’s result tables, and no figure in this paper was created or altered with generative AI. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is operational air traffic data held by EUROCONTROL and processed under existing data-handling arrangements; it cannot be made publicly available. Aggregated tables underlying every figure are available from the corresponding author upon reasonable request, in the de-identified form this paper itself uses: operators, city pairs and waypoints appear in them by rank rather than by name, which leaves every number and every ordering unchanged. Per-flight tables are excluded from the release.

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